

Driving with Algorithms Beyond Gig Work: Investigating How Algorithmic Management Affects Workers' Practices in an On-Demand Ride-Pooling Service

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Abstract

On-demand ride-pooling (ODRP) services are a new alternative modes of transportation that have recently emerged due to technological advancements. While algorithmic management plays a crucial role in ODRP services and can create complex workplace dynamics, the experiences of ODRP workers remain underexplored in the HCI field. To address this gap, we interviewed 16 drivers of Shucle, an ODRP service in South Korea. We examined the drivers' detailed work practices, focusing on the perceived challenges of working under algorithmic management and the perceived benefits and necessity of algorithmic management. This paper provides empirical evidence of the impact of algorithmic management on ODRP drivers' work environments and discusses the implications of our findings for supporting algorithmic workplaces in ODRP services. By positioning ODRP drivers as company employees embedded within a vast, dynamic traffic environment, our study extends algorithmic management scholarship beyond gig work and other algorithmic work contexts, offering fresh insights into how autonomy and accountability are configured across algorithmic workplaces.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Algorithmic management, On-demand ride-pooling services, Worker-centered design, Worker practice, Work experience, On demand work, Labor issues, Work assignment, Performance evaluation, Human-centered algorithm, Sharing economy, Shared mobility, Demand-responsive transit

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1 INTRODUCTION

Recent technological advances have led to a new smart mobility service, the **On-Demand Ride-Pooling (ODRP) service**. An ODRP service is a transportation system that enables multiple passengers to share a single vehicle for their journeys. This service differs from traditional public transportation in that it has no fixed routes or stops and instead dynamically adjusts vehicle dispatch, routes, and stops through real-time algorithmic decision-making, enabling passengers with different origins and destinations to share rides. ODRP services are emerging as a promising new mode of transportation that can alleviate urban traffic congestion and help reduce environmental pollution [11, 44, 77].

ODRP service companies aim to enable efficient service operation by increasing vehicle occupancy rates and decreasing driving distances by gathering passengers with similar travel routes into a single vehicle [3]. To achieve these goals, companies use **algorithmic management** in the workplace. Drivers, i.e., workers of the ODRP services, are assigned passengers through the GPS screen based on location-based algorithmic decision-making, and they are directed on the route to follow and required to drive the vehicle accordingly. Specifically, drivers are directed to arrive at the pick-up point in time for the scheduled passenger boarding time and are assigned work hours, break times, and rest stops. Furthermore, algorithmic systems monitor and evaluate drivers to ensure that they perform the assigned tasks correctly.

Previous studies have shown that adopting algorithmic management in the workplace can increase worker productivity and create efficient and fair work processes by automating tasks humans cannot perform and providing data-driven decisions [9, 18, 84]. However, research also highlights that algorithmic management can lead



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to unexpected effects on worker practices and create complex workplace dynamics [5, 42, 73]. Recent literature has emphasized the negative aspects of algorithmic management on workers, contrary to companies' idealized expectations, and reports that these effects are manifested across various domains and services [1, 21, 36].

The work practices of ODRP drivers are unique in that their tasks dynamically change based on the real-time demands of service users, and they perform tasks in a broad traffic environment. Therefore, worker practices in ODRP services may exhibit nuanced features distinct from those of gig work (e.g., ridesharing [41, 82], food delivery services [29, 87]) or stationary workplaces (e.g., hospitality industry [79], factory [19], and warehouse [20]). When employing algorithmic management, it is crucial to investigate which worker practices arise within the inherent characteristics of new services and work environments [46, 50]. However, despite the expansion of ODRP services [2, 22], the experiences of ODRP workers remain underexplored in the HCI field.

Given this background, this study aimed to investigate how algorithmic management in ODRP services affects worker practices. Specifically, we aimed to answer the following research questions:

- **RQ 1:** What are the perceived challenges that ODRP drivers encounter under algorithmic management in ODRP work environments?
- **RQ 2:** What are the perceived benefits and necessities that ODRP drivers associate with algorithmic management in ODRP work environments?

To address our research questions, we conducted semi-structured interviews with 16 drivers employed by *Shucle*, an ODRP service in South Korea. This study examines how algorithmic management shapes work practices in dynamic service environments by highlighting both the challenges it imposes on workers (RQ1)—such as drivers' neglected discretion, need for human judgment, compliance pressures, and misattributed accountability—and the benefits it offers (RQ2), including reduced work burdens, efficient workflows, and enhanced fairness. By synthesizing the findings from RQ1 and RQ2, this research contributes to the HCI community by deepening our understanding of algorithmic work environments within ODRP services and offering design implications to promote work practices that balance operational efficiency with human intervention and autonomy.

2 BACKGROUND AND RELATED WORK

In this section, we first present an overview of how ODRP services work and the role of algorithms in the services. Next, we provide a review of previous work on the impact of algorithmic management on workers, as well as their perceptions and responses to it. Lastly, we compare the workplaces of ODRP services with those in other domains that use algorithmic management, highlighting the unique characteristics of the ODRP work environment.

2.1 On-Demand Ride-Pooling Services

On-demand ride-pooling (ODRP) services are transportation systems that enable multiple passengers to share a single vehicle for their journeys, with routes adjusted in real-time based on passenger demands. These services are operated by a variety of companies

around the world, such as *Shucle* [75], *Moia* [59], *Via* [83], *Uber-Pool* [81], and *Lyftline* [52]. These ODRP services are distinguished from conventional transportation, such as taxis and buses, because they rely on algorithmic systems to manage and determine most operational elements instead of relying on human drivers.

ODRP services are typically operated by companies, and their goal is to improve the efficiency of vehicle operation by grouping passengers traveling similar routes, which reduces driving distance and increases vehicle occupancy [12]. In other words, ODRP services emphasize the overall efficiency of the passenger group, functioning as a resource-sharing service designed for multiple users. To achieve this, algorithmic systems match passengers with similar origins and destinations, dispatch the most appropriate vehicles, generate optimal routes, and adjust routes in response to real-time demand from service users [3]. Additionally, these systems manage boarding and disembarking orders and assign passenger pick-up and drop-off points [31, 78]. Passenger pick-up and drop-off points are assigned by the algorithms to one of several locations near the passenger's set origins and destinations, considering the vehicle's overall route and the origins and destinations of other passengers [31]. Passengers can request a vehicle through the mobile app and set origins and destinations. Upon dispatching the vehicle, they can check the estimated time of pick-up and arrival through the app. In addition, the algorithmic systems analyze the real-time travel demand along with the vehicle's location-based data and predict future demand based on the data [35, 38, 47]. Overall, this algorithmic system coordinates and optimizes all aspects of vehicle operation, adjusting routes in real-time according to passengers' demand and enabling multiple passengers to share a vehicle efficiently [89].

Since ODRP services operate under centralized algorithmic systems, ODRP drivers need to perform tasks directed by algorithms. They receive information such as driving routes, passenger details, and pick-up and drop-off locations generated by the algorithms through the GPS screen in the vehicle and drive accordingly. Moreover, their driving routes and task adherence are recorded, and these records are used for monitoring and evaluating their work. Such algorithmic management in ODRP services can have complex effects on work practices; in particular, ODRP drivers may face new challenges they have not experienced in their conventional driving.

2.2 Algorithmic Management and its Impact on Workers

In the HCI field, algorithmic management's impact on workers has become a significant research area. The specific functions of algorithmic management vary according to the work context, but algorithms typically include directing workers to their tasks and assigning them work schedules. Additionally, algorithms evaluate workers' performance by recording, monitoring, and judging their tasks.

Using algorithmic management can improve workplace fairness and efficiency by automating routine tasks and providing data-driven decision-making [9, 84]. For this reason, many companies adopt algorithmic management in their workplaces to enhance worker productivity and oversee optimization on a large scale, ultimately increasing profits [18]. In addition, algorithmic management can reduce workers' workloads and create opportunities for worker

self-improvement and career advancement [51]. Moreover, algorithms' predictable and systematic work direction and evaluation can decrease workers' stress [9].

However, despite these potential benefits, many previous studies reveal that algorithmic management changes relations among employers, platforms, and workers and shapes complex work practices. In particular, algorithmic management can negatively impact work practices by giving employers and platforms more control and power [69]. Additionally, it could control worker behaviors through surveillance for task compliance, algorithmic opacity, behavioral nudges, and rigid evaluation measures [5, 48].

Increased control and monitoring by algorithmic systems could reduce workers' discretion [36]. Recent literature indicates that workers' autonomy and discretion are undermined, leading to burnout, anxiety, and depression [14, 26]. Moreover, workers could be subject to compliance pressures and concerns about privacy violations [48, 68], as their work performance is monitored in real-time in various ways [23]. This persistent surveillance, which may carry significant consequences, such as layoffs, is often perceived as hostile by workers [65]. In addition, this surveillance can be further intensified by the opacity of the algorithmic systems [10, 17, 28]. Workers often struggle to comprehend the operational processes of algorithmic systems, leading to confusion about the reasons for following algorithmic directions or the potential consequences of non-compliance [49, 64, 67]. This lack of transparency may compel workers to adhere rigidly to algorithmic directions [90] or, conversely, foster reluctance to comply [53].

Furthermore, workers may experience asymmetries in work practices due to algorithmic management. Workers have limited authority to ignore work directions due to information asymmetry, such as the inability to access the customer's location or delivery fare [41]. This asymmetry also limits workers' autonomy, undermining their capacity to make informed decisions [82]. On the other hand, employers or workplace supervisors may have more control over algorithmic management through increased access to information (e.g., ratings), workflows, and worker choices [40]. Due to the unilateral flow of information and limited communication between workers and employers, workers often struggle to understand how their performance is evaluated or the reasons behind those evaluations [32]. Thus, workers describe algorithmic systems as bosses [58], expressing frustration toward algorithmic management but feeling powerless against employers and giant companies [43]. Furthermore, uncertainty stemming from information asymmetry and algorithmic opacity can impede collective action among workers [88]. This obstruction of collective action hinders the establishment of improved work practices under algorithmic management.

In addition, service users can evaluate workers' performance through reviews, ratings, and rankings. Based on these evaluations, algorithmic management may restrict certain tasks for workers with low user ratings [34, 56, 69]. This creates a persistent sense of unease among workers, reinforcing the perception that all service users act as middle managers [62]. For instance, a study on Uber drivers revealed that they experience constant pressure to meet behavioral expectations (e.g., avoiding mobile phone use while driving) and maintain minimum customer rating thresholds, resulting in significant work-related stress [57, 65].

In summary, the impact of algorithmic management on workers poses multifaceted challenges. This study aims to expand the existing body of research on algorithmic management within emerging services by exploring the relatively underexplored work practices of ODRP drivers operating such systems. Specifically, our research addresses these challenges by focusing on the distinctive work practices observed in the context of ODRP services.

2.3 ODRP Services and its Work Environment with Algorithmic Management

This section examines how the work environment for ODRP drivers, who work with algorithmic systems, differs from that in other domains, highlighting the unique characteristics of the ODRP work environment. First, ODRP drivers employed by a company are influenced not only by algorithmic systems but also by stakeholders within the company, resulting in more complex dynamics. Since ODRP services are operated by companies, drivers cannot work independently, as vehicles rely on centralized dispatching and scheduling for efficient route management. Thus, the work environment of ODRP drivers, as company employees, shares the work context of both gig workers and traditional workplaces. Additionally, traditional workplaces that have adopted algorithmic management (e.g., warehouses [19, 27] and hotels [61, 79]) share commonalities with ODRP work environments. In both contexts, workers are employed by companies, perform tasks directed by algorithmic systems, and are supervised by organizational leaders or local managers based on algorithmic evaluation outcomes [27, 63, 86]. However, the scale of the workplace where algorithmic systems are applied to workers can introduce a significant distinction. ODRP drivers operate in vast, dynamic traffic environments where external variables are less predictable compared to the controlled, confined settings of warehouses or hotels. This broader scope of operation can make the ODRP work environment uniquely challenging, as it is subject to a broader array of unexpected factors beyond the company's control.

Another relevant comparison is with location-based gig workers, such as ridesharing (e.g., Uber, Lyft) and food delivery (e.g., Deliveroo, Foodora) drivers. Like ODRP drivers, these gig workers also perform tasks directed by algorithms in broad traffic environments rather than in a confined and limited workplace [37, 41, 82]. They are also monitored for task execution time, compliance with algorithmic directions, and performance evaluations from service users [37, 41, 49, 80]. Based on these evaluations, if a worker's performance falls below a threshold, they are penalized by, for example, not assigning the worker a task, leading to increased pressure on them to perform [41, 69, 74]. However, unlike ODRP drivers, they can theoretically have the option to choose certain tasks and refuse to follow directions without facing additional detriment from an employer, as they are not formally employed by a company (i.e., independent contractors) [66]. Moreover, ridesharing drivers can exercise flexibility and choice in their work—for example, by turning the driver app on for the daily commute or turning off the driver mode when returning from a long-ride—and their experiential knowledge, along with learned and improvisational strategies, play a critical role in social sensemaking within algorithmically managed work environments [37, 49]. Therefore, the ODRP work

environment is distinct in that drivers must follow algorithmic directions with almost no task flexibility, unlike gig workers.

In addition, gig workers are motivated by piece wages, where compensation is directly tied to productivity. In contrast, ODRP drivers employed by companies receive hourly wages and may have explicit managerial oversight. The piece wage system, while potentially exploitative, minimizes the need for micromanagement by incentivizing productivity through financial rewards [4, 54]. As a result, ODRP drivers are more likely to experience explicit disciplinary controls, reflecting a more hierarchical and supervised work environment.

Another relevant comparison can be made with parcel delivery workers, such as those employed by Amazon and UPS, who share similarities with ODRP drivers. Both groups follow algorithmically generated work directions (e.g., delivery routes) and operate in broad traffic environments [63]. However, parcel delivery workers are less influenced by service stakeholders, as their tasks typically do not involve direct interaction with end-users, and their instructions are fixed at the outset. In contrast, ODRP drivers' workflows are subject to end-user demands, introducing additional variability and complexity.

Taken together, the work environment of ODRP drivers exhibits unique characteristics that differentiate it from other algorithmic workplaces previously studied. The intersection of employment status, dynamic traffic environments, and stakeholder interactions creates nuanced work practices distinct from previously studied contexts. This study focuses on the relatively underexplored domain of ODRP services, investigating how ODRP drivers navigate algorithmic management within these work environments.

3 METHODS

3.1 Research Context: *Shucle*, an ODRP Service in South Korea

We first introduce the research context of our study: *Shucle*, an ODRP service operated in South Korea by Hyundai Motor Group [75]¹. *Shucle* has been piloted in Sejong and Paju Cities in South Korea since 2021 with special permission from the government for demonstration, and the service is currently expanding to various cities. We selected *Shucle* as the research context for a typical ODRP service because it is the first ODRP service to be launched and operated in various cities across South Korea. (To our knowledge, *Shucle* services operate consistently across cities, with no significant differences in terms of pricing, routes, or operational management.) *Shucle* is also the most used ODRP service in South Korea [76]. We believe that *Shucle*'s maturity can effectively reveal the worker practices under algorithmic management, as it may face fewer challenges related to technical implementation and administrative operations compared to other ODRP services. In addition, the maximum number of passengers allowed on a *Shucle* vehicle is ten passengers. *Shucle* employs approximately 80 drivers, roughly 35 of whom work in Paju City. In each city, the *Shucle* service employs one or two field managers who are responsible for supervising the drivers. These managers oversee drivers by aggregating

work performance data from algorithmic systems and ratings from service users.

Although our study focuses on a single case (*Shucle*), it is noteworthy that ODRP services are emerging globally, with companies such as *Shucle*, *Moia*, *Via*, *UberPOOL*, and *BRIDJ* also offering similar ODRP services [25, 55, 70]. Preliminary evidence suggests that the underlying algorithms and operational models—such as those used for high-capacity vehicles (8+ passengers)—are similar between *Shucle* and *Moia* [85, 91]. Finally, we conducted this study independently, with no affiliation or financial relationship with *Shucle* or its operating company.

3.2 Participants Recruitment

First, we contacted *Shucle*'s field managers in each service area (Paju, Sejong) and explained the purpose of our study. After receiving their consent, we requested them to post participant recruitment posts on their bulletin boards at their depots and to share posts in group chats where all the drivers were involved. We also requested that recruited participants introduce their fellow drivers.

As a result, we recruited 16 participants (5 in Sejong and 11 in Paju). The average age of the participants is 57.8 years (SD=6.0, MIN=50, MAX=66), with 12 males and 4 females. The average experience of drivers at *Shucle* at the time of the interview was 15.3 months (SD=5.7, MIN=4, MAX=22). It was 22 months after the first launch in Sejong and 18 months in Paju at the time of the interviews.

All of the participants were previously employed in the transportation industry—working in diverse roles such as taxis, city buses, express, school buses, and airport limousines—which made them seasoned professionals with extensive driving experience. Many had reached or were nearing retirement in their former careers, contributing to the higher average age observed in our sample. Because *Shucle* hires drivers on a contractual basis, without the constraints of Korean traditional retirement policies, these experienced professionals continue working regardless of age in *Shucle*. In addition, most participants were local residents of the service areas, further emphasizing the veteran character of this group.

3.3 Study Procedures

We first sought to identify and organize the role of algorithmic management in the *Shucle* service (see Table 1). Based on multiple *Shucle* rides by researchers and initial feedback from *Shucle* drivers and a field manager, we organized this table as a preparatory step before conducting the interviews, which focused on exploring participants' work practices in depth. We did not include the interview with the field manager in our data analysis. However, he helped us structure the content of Table 1 and confirmed the specific situation in the table. Through this table, we aim to provide readers with an overview of participants' tasks, facilitating a better understanding of the newly emerging ODRP service. A typical day begins with receiving an initial passenger dispatch from the depot. Throughout the shift, drivers continuously follow algorithmic directions to pick-up and drop-off passengers (Situations <1> to <4>), based on requests placed by passengers through the app. Even when no passengers are onboard, the system guides vehicles toward predicted high-demand areas (Situations <5> and <6>). While driving, drivers

¹<https://www.shucle.com/>

are continuously monitored to ensure route compliance (Situation <7>). Scheduled breaks are algorithmically assigned at designated rest stops (Situation <8>), and at the end of the shift, drivers follow system-generated routes back to the depot (Situation <9>). We also referenced each role as 'Situation <#>' in §4 to help the readers' understanding.

We also used a table to organize the questions for the semi-structured interviews and to serve as a resource for generating follow-up questions. We first began the semi-structured interviews by asking participants about their work experience with *Shucle* services and their experience working in other modes of transportation. We then asked participants to describe their typical work routines and practices working on *Shucle* services (e.g., *What does your typical work day look like?*). Next, we asked them to elaborate on their understanding and perception of algorithmic management (e.g., *Do you understand the algorithmic system of Shucle, and what are your perceptions of it?*), their satisfaction with current work experience (e.g., *Are you satisfied with working for Shucle services?*), and expectations (e.g., *What would you like to improve about the current work practice, and how?*). Additional questions were flexibly added, depending on participants' responses in the questionnaire for recruitment and interesting comments made during the interviews.

All interview sessions that addressed this study lasted approximately 40 and 60 minutes, with the majority of them conducted online using Zoom. For the five participants who had difficulty using Zoom, we used a conference room near their workplace to conduct in-person interviews. Each interview was conducted individually. All the online and in-person interviews were recorded and transcribed. Participants were paid 50,000 KRW (approx. 37 USD) each for their participation.

3.4 Ethical Considerations

Before the study, participants were informed of the purpose of our research, their rights, and how their data would be processed and protected. They were also informed that if they felt uncomfortable, they could choose not to answer any questions or withdraw participation at any time. In addition, we clearly stated that, as researchers, we had no interest in *Shucle* and guaranteed that participants' data and information would never be shared with *Shucle*. The first author received written consent from each participant before starting an interview.

In this study, we used participant numbers D1 to D16 when referring to them. We also decided not to provide detailed demographics to ensure the anonymity of the participants. This is because a small number of drivers are employed by *Shucle*, making it possible to infer the identity of participants based on their information, such as age, gender, and experience. This study was approved by the Institutional Review Board (IRB No. KH2022-199).

3.5 Data Analysis

The transcribed data from the interviews were analyzed using thematic analysis [15], providing flexibility and reflexivity in examining qualitative data [16]. First, three researchers corrected the transcriptions while referencing the recordings. After verifying the transcriptions, we conducted open coding using ATLAS.ti [6].

Through iterative discussion, we cross-checked and developed the codes until we reached a consensus. The analysis focused on identifying patterns of ODRP drivers' current work practices under algorithmic management and their perceptions of algorithmic management. As a result, we organized two primary themes and six sub-themes presented in § 4.

4 FINDINGS

Our study uncovered the work practices of ODRP drivers under the influence of algorithmic management. Our key findings were the participants' perceived challenges (§4.1; RQ1) and the perceived benefits and necessity (§4.2; RQ2) of algorithmic management in *Shucle*. We report the details of our findings below.

4.1 ODRP Drivers' Perceived Challenges to Algorithmic Management

First, we found that participants encountered various challenges associated with algorithmic management during their work. Specifically, they experienced the pressure of unceasing passenger demands combined with the neglect of drivers' discretion (§4.1.1). Participants also expressed concerns about algorithmic misalignments with real-world traffic and highlighted the need for human judgment (§4.1.2). Additionally, they highlighted the inability of algorithmic worker monitoring to capture situational contexts, thereby increasing compliance pressures (§4.1.3). Finally, participants reported feeling disempowered by algorithmic routing decisions, as these often resulted in misattributed accountability for passenger complaints (§4.1.4).

4.1.1 Unceasing Passenger Demands and Neglected Drivers' Discretion. *Shucle* drivers, unlike gig workers, are not independent contractors, so they cannot selectively accept passenger calls. Participants had to respond to passenger demand according to the dispatch of algorithms. This unilateral dispatch system caused significant *"physical and mental exhaustion"* (D8). They felt burdened by the constant obligation to comply with algorithmic dispatches that disregarded their individual circumstances, as illustrated by D7:

"Passenger calls keep coming in, and we can't stop them. [...] So I just kept holding on, and finally, when there were no passengers, we found the nearest restroom in the service area and went there." (D7)

Although these dispatches are essential to ensure a smooth service experience for passengers, participants said the algorithms should consider *"the circumstances of humans, which are not mechanical"* (D8). Also, D15 wanted to bring *"humanism"* to the current algorithms, noting that *"Just like we try to roll with the algorithm's sudden route changes, I wish the algorithm could also take into account the unpredictable, unavoidable physiological needs we humans have."* D15 explained that while drivers are required to constantly respond to the unpredictable demands of passengers, drivers' unpredictable situations, such as their physiological needs, are not adequately considered.

Moreover, participants were directed to quickly reposition their vehicles according to algorithmic directions to prepare for potential passenger demand, even when there was no actual demand. For

Table 1: The role of algorithmic management in Shucle.

No.	Situation
<1>	Passengers can set their personal destination and origin through the app, and the algorithms assign passenger pick-up and drop-off points in the service area.
<2>	The vehicle is dispatched to pick up the requested passengers, and the algorithms generate a route based on their pick-up and drop-off points.
<3>	The driver picks up the passenger after arriving at the assigned pick-up point; if the passenger has not yet arrived, the algorithms direct the driver to wait only 1 minute to avoid delays.
<4>	If a new passenger call is dispatched on the way to the next scheduled drop-off or pick-up point, the existing route is changed, and the driver drives the vehicle along the changed route. This may delay the scheduled arrival and boarding times, change the arrival and boarding order, and alter the drop-off and pick-up points for passengers.
<5>	The algorithms can route vehicles to points where passenger demand is predicted based on previous call data, even if there are no new passenger calls and no passengers on board. The purpose of this algorithm is to respond quickly when a new passenger call occurs, maximize profitability, and distribute vehicles evenly in a service area.
<6>	Even with passengers onboard, the algorithms may direct routes through points where passenger demand is predicted based on previous call data rather than driving the shortest distance to the destination.
<7>	The algorithmic systems monitor the drivers to ensure that they follow the route as directed and do not stop for more than 5 min along the way.
<8>	After the morning/afternoon shift, drivers take an assigned break (including a lunch/dinner break); the algorithms direct drivers to one of several rest stops in the service area. During the break, drivers park their vehicles at the rest stop and take a break nearby.
<9>	After the break, the preceding tasks are repeated, and the algorithms generate a route back to the depot at the end of the shift.

example, they were directed to drive to a specific rest stop in the service area for break time, which was based on passenger considerations rather than their preferences (e.g., Table 1, Situation <8>). Some participants described scenarios where they were directed to rest stops with few nearby restaurants, which prevented them from having a proper meal. Such experiences led participants to complain about the inability to take quality breaks:

"The AI even directs us where to eat. It practically determines where and what we are going to eat." (D14)

"There was a rest stop near where I was driving, but algorithms sent me to another rest stop in the area where there were more potential passengers. The problem is, there's nowhere to eat around there. [...] I have to follow whatever the AI system says, and no questions are asked. I just don't get why it considers passengers even during our break time instead of thinking about us." (D15)

Collectively, the participants expressed dissatisfaction with the algorithmic system that prioritized passenger requests over drivers' needs. Similarly to other workers, ODRP drivers operate with algorithmic directions; however, the requirement to strictly adhere to passenger demands without personal discretion—stemming from the integration of algorithmic control and a passenger-centric service model—creates distinct work practices that differentiate them from workers in other algorithmic management contexts. This lack of control over passenger demand, combined with the rigidity of algorithmic decisions, contributed to their sense of frustration and reduced job satisfaction.

4.1.2 Algorithmic Misalignments with Real-world Traffic and Drivers' Need for Human Judgment. Participants mentioned situations where they encountered algorithmic directions that did not consider real-world traffic environments. They reported that they are often directed onto narrow or obstructed roads because *Shucle's* algorithmic systems were unable to recognize the physical surroundings of the vehicle: *"There was a time when some roads were closed due to a local event like a concert. But I was instructed to drive on that road" (D1)*. Additionally, some participants (D4, D5) recalled the scenarios in which they encountered directions that directed the vehicle to make a turn when the distance between the stop and the turning point was too short to change lanes safely.

Participants also recalled situations where they faced a sudden change in their existing route when new passenger demands arose (e.g., Table 1, Situation <4>). However, they said that it was difficult to respond immediately because these algorithmic directions were delivered instantly without prior notice, as described by D6 and D8:

"We're being guided straight at first. So, we keep going straight, and just as we get close to an intersection, it suddenly directs us to make a left turn. In this situation, we can't follow the direction." (D6)

"When I drive for a long time, I get tired. In those situations, I've often been in situations where I was getting ready to turn right when I was directed to do so, but the direction suddenly changed to a left turn. I've almost had accidents in such situations." (D8)

Participants (D1, D7, D9) believed that their familiarity with the service area's traffic conditions and their ability to adapt flexibly would help address such issues: *"To meet a new passenger's scheduled pick-up time, the vehicle is often unexpectedly rerouted. But I think sometimes it's better to judge the traffic condition and adjust the vehicle's speed appropriately rather than change the route. I know the local geography and traffic conditions here, so we can make adjustments without causing problems"* (D7). However, these participants pointed out that the current system does not allow them any discretionary decision-making and expressed expectations for more collaborative systems that incorporate human drivers' competencies. The following quotes from D13, D1, and D9 illustrate this expectation:

"The problem of routes suddenly changing is something drivers just have to deal with depending on the situation. [...] In such situations, the driver's judgment is most important." (D13)

"The current system is pretty good, but in specific situations, it's not about solving it with the algorithms but about the driver intervening. There's a role for the driver, and the system should acknowledge our ability and assign tasks to us." (D1)

"I believe that the algorithm doesn't need to do everything 100%. It's good enough if it does 90% to 95%. We can fill in the remaining. [...] Human drivers have our own strengths." (D9)

In summary, participants expressed frustration with algorithmic directions that disregarded real-world traffic conditions, highlighting the need for collaborative systems. While location-based gig workers also operate under algorithmic directions in traffic environments with some flexibility, ODRP drivers face more rigid constraints, as they are required to adhere strictly to these directions. They believed that incorporating human judgment alongside algorithms could mitigate these challenges and enhance system efficiency.

4.1.3 Failing to Capture Context in Algorithmic Worker-Monitoring and Drivers' Compliance Pressures. Although participants often received directions that were difficult to execute or seemed unreasonable, they attempted to follow them whenever possible unless doing so posed a safety hazard. Participants explained that this compliance stemmed from concerns about being evaluated negatively if they did not strictly adhere to algorithmic directions, as D3 illustrated:

"There was an accident on the road I was directed to drive, so traffic was congested. I would have taken a detour if I were driving alone in this situation. However, I would be the only one penalized if I were recorded as out of route, so I had no choice but to follow the existing route, even though it would delay my passengers' arrival time." (D3)

As drivers work alone without field managers physically present, monitoring and evaluating workers is necessary for the operational management of ODRP services. To address this, *Shucle's* algorithmic systems detect driving records such as drivers' off-route, stops over 5 minutes, and sudden acceleration or deceleration. These

records are then sent to the field manager responsible for the overall supervision of workers. However, participants perceived that field managers often evaluated their performance primarily based on numerical data. As D9 described, this reliance on *"binary criteria"* meant that field managers focused solely on whether drivers met algorithmically defined thresholds, disregarding situational factors. They believed this approach made it difficult to consider the context and specific situations in which these records were generated:

"If a lot of route deviations are detected, they (field managers) just focus on 'why are there so many deviations?' So we try really hard not to deviate from the route as much as possible when driving." (D11)

Similarly, D10 emphasized that algorithmic interpretations of the situations often failed to understand the reasons behind their deviations accurately:

"(Recalling the time when they were pointed out by the field manager for a '5-minute stop') During rush hours, traffic gets congested. If we have to wait for signals, let's say three times or so, vehicles hardly move during that time, right? But it seems like the system judges that I took a break on my own." (D10)

"They (field managers) should check the dashcam to understand the situation. [...] They are reprimanding us based only on what the system detected." (D13)

As D13 mentioned, participants wanted to use dashcam records to be reviewed for vindication. However, the large amount of data detected by algorithms made this process time-consuming and rarely feasible in practice. Moreover, since the dashcams were provided by the service company, participants found it challenging to access the footage independently, thereby limiting their ability to provide evidence or challenge inaccurate evaluations. To address these challenges, some participants took proactive steps to document contextual evidence to justify their performance, underscoring the inadequacies of the current monitoring system and the drivers' need to compensate for its shortcomings. D6's experience well describes these behaviors:

"I sometimes receive the system's driving instructions that are impossible to follow. For example, in narrow alleys where only one car can pass, the Shucle vehicle's body is large, so it can't pass. In such cases, I take out my phone, video-record the surroundings, and make notes. I record why I deviated from the route, the time and place, and even the detailed situation. I jot down everything as I need to have evidence." (D6)

Additionally, participants sought to ensure that their feedback on improving work practices was incorporated into *Shucle's* algorithmic systems. For example, they shared issues in a *Shucle* drivers' group chat room (D10, D11, D14), reported cases when service developers came to their workplace occasionally (D6, D7, D14, D15, D16), or contacted customer service to report issues (D6, D7).

In summary, the participants were consistently aware that their work performance was being monitored by algorithmic systems and field managers. This hybrid management introduces unique challenges and imposes additional constraints on ODRP drivers' work practices. However, the lack of contextual understanding in

these systems often left participants needing to justify their actions to field managers, highlighting the need for systems that better integrate contextual insights.

4.1.4 Passenger Frustrations from Algorithmic Decisions and Drivers' Misattributed Accountability. As implied in the previous sections, participants often expressed frustration over having to follow algorithmic decisions throughout their working hours. Many participants (9 out of 16) perceived themselves as mere agents who must follow routes generated by algorithmic systems, however, passengers often misunderstood that drivers were making their own driving decisions. Due to the nature of ODRP services, multiple passengers share a single vehicle, making it impossible for all passengers to take the shortest route to their destinations. Additionally, new passenger calls could delay scheduled arrival times and change the order of disembarkation [78] (e.g., Table 1, Situation <4, 6>). Although these situations were caused by algorithms, the drivers—who directly interacted with passengers—often bore the brunt of their complaints. D5's interview illustrates this experience.

"I still remember that passenger's ID... He made me feel humiliated, yelling at me and asking why I was making a detour with a left turn when I could just go straight. [...] I was just driving according to the algorithm. He was in the backseat blaming me for no flexibility, and it got me mad." (D5)

When passengers experienced delays, they often expressed dissatisfaction by rating the service poorly through the app after disembarking. While passengers express low ratings for legitimate issues, such as drivers slacking off or driving recklessly, time delays caused by algorithmic decisions are an entirely separate matter. These delays stem from the algorithms' decision-making, yet participants expressed frustration over receiving mixed ratings from passengers.

"Passengers don't know much about how the [Shucle's] AI system operates, so they understand it as if I, the driver, changed the drop-off location on my own. They get angry with me and engage in rating attacks even if I explain." (D6) (e.g., Table 1, Situation <4>)

"I used to be the top-rated driver at my previous job (city bus) because I provided good service to passengers and drove safely. But none of that matters with this rating system. Ratings just go down if the passengers arrive late, or if the route is detoured, then they give low ratings." (D15)

These ratings are then delivered to the Shucle's field manager, who oversees drivers in the service area. However, participants noted that field managers primarily rely on the average star ratings as the key personnel evaluation metric, which only reflects numerical information without context. As a result, drivers constantly worry about rating drops caused entirely by algorithmic decision-making, and this even makes them feel anxious about their job security.

Despite these challenges, participants expressed a desire to play a more active role in providing better service to passengers. They believed they could provide better service in certain situations if their

knowledge and judgment were utilized. However, they were frustrated that such actions were not officially permitted. D11 shared an example of this frustration:

"I know the geography of this neighborhood like the back of my hand, and I think we know it better than the AI. We know which routes are shorter and faster. We just follow the AI's instructions, but when we do, the passengers' ETA keeps getting delayed. It's frustrating because the AI gives directions that are different from what I think is right. If our judgments were respected, things would run more smoothly, and passengers could arrive even a minute or two earlier, but it's a shame that it's not like that." (D11)

This rigid operational framework, when coupled with the workplace surveillance mechanisms detailed in §4.1.3, significantly undermined drivers' motivation to deliver better service to service users. Consequently, they began to regard their work not as an opportunity to excel, but as something to be endured until the workday was over.

"I don't know where to find the motivation to work harder anymore. I used to be passionate about arriving faster to pick up one more passenger, but now I just think I'm killing my time until my shift ends." (D7)

In summary, participants expressed frustration at being perceived as mere agents of algorithmic decisions, often facing misunderstandings and complaints from passengers. The work environment of ODRP drivers, who interact directly with passengers within the vehicle, requires them to handle complaints about delays and detours caused by algorithmic routing. This misattribution of responsibility not only undermines their ratings but also affects their job security, professional identity, and employee motivation, underscoring the tension between human judgment and algorithmic control in ODRP services.

4.2 ODRP Drivers' Perceived Benefit and Necessity of Algorithmic Management

While §4.1 outlined the challenges drivers face under algorithmic management, it is important to acknowledge that they also recognize its benefits and necessity. Despite these challenges, we observed that algorithmic systems in Shucle could mitigate their incidental work burden (§4.2.1) and create a fair work environment (§4.2.2).

4.2.1 Mitigating Work Burden. Overall, many participants (7 out of 16) acknowledged the necessity of Shucle's algorithmic systems for generating optimal routes in response to passenger demand, which complements the abilities of human workers: *"I often praise the routes that AI creates. I think AI can only organize the overall routes so that many passengers can share a vehicle efficiently and get passengers on and off at the right places. I think this function is essential to our work, and if humans organized this, the routes would likely be tangled and inefficient"* (D15). Some participants (D3, D12) mentioned that this automation ensured their work productivity, allowing them to focus on driving without additional cognitive effort.

Additionally, some participants (D2, D9) noted that incorporating algorithmic systems made a non-competitive service model possible, where they only needed to perform the tasks assigned to them:

"In Shucle, unlike taxis, I only have to follow the directions when driving, so I feel less pressure about work performance. In taxis, I have to compete with other drivers to get passenger calls, but in Shucle, I don't have to worry about finding passengers. So I feel relatively less anxious." (D2)

In summary, incorporating algorithmic systems in *Shucle* allowed participants to focus on their primary task of driving by alleviating the burden of incidental work. Their positive perspectives underscore the potential of algorithmic systems to complement human skills and enable structured, efficient workflows.

4.2.2 Creating a Fair Work Environment. In addition, some participants (D3, D4, D6) argued that, despite some drawbacks, algorithmic monitoring is necessary to manage workers who may exploit the system. While they acknowledged that their workplaces may seem strict, they pointed out that if workers are not effectively controlled or given excessive flexibility, some drivers may exploit the algorithmic systems:

"I think that if drivers are given too much flexibility in their work without systematic management, some drivers might abuse it. This actually happened in the early days of the service. At that time, the system didn't monitor the drivers' '5-minute stop'. So, three veteran drivers got together and rested for almost an hour without driving. I think this can happen again if there is no monitoring." (D6)

D3 also mentioned covertly shared deviant behavior among *Shucle* drivers: if drivers intentionally drove slowly, the algorithms reduce the number of dispatches assigned to that vehicle, allowing them to drive more comfortably with fewer passengers. D3 said that they have witnessed drivers intentionally driving slowly and pointed out that this deviant behavior can increase the burden on honest drivers: *"Normally, I drive about 100 kilometers a day, but when working with slacking drivers (who drive deliberately slowly), it causes us to even drive up to 120 kilometers."* D3 further explained that the algorithmic system does not assign additional passengers to drivers who are intentionally driving slowly (i.e., vehicles with multiple passengers waiting to board). As a result, new passengers are assigned to other drivers, which increases those drivers' workload and driving distance. D4 also commented on these deviant drivers and argued that worker monitoring and evaluation by algorithms should be more detailed and strict.²

As such, participants noted the necessity of algorithmic management to foster a fair work environment in the ODRP services.

²We did not have access to *Shucle*'s system data, so we lack direct evidence that drivers are indeed assigned fewer passenger calls if they intentionally drive slowly. However, in our informal interview with the field manager, it was confirmed that situations in which there is a significant difference in the number of kilometers driven per day between workers often occur.

5 DISCUSSION & DESIGN IMPLICATIONS

Reflecting on our findings, participants perceived that in *Shucle*, algorithmic management reduced their workload by providing optimized routes and real-time task assignments while ensuring work efficiency and fairness. These perceived benefits and the necessity of algorithmic management align with previous studies [49, 64]. Thus, algorithmic management effectively fulfilled its fundamental role within the ODRP work environment.

Nevertheless, participants frequently encountered unconventional challenges arising from the unique nature of ODRP workplaces. To address these challenges, this section discusses design implications for supporting algorithmic work practices in ODRP services.

5.1 Towards More Flexible ODRP Workplaces that Embrace Human Workers' Autonomy

The work environment of ODRP services is the broader traffic system, which—unlike highly controlled workplaces such as warehouses [19] designed for algorithmic management—presents many unpredictable external factors. In this unique workplace, ODRP drivers perceived their autonomy and flexibility to be limited. Their roles were confined to following algorithmic directions without any additional authority or discretion. This rigid system may stem from the belief held by ODRP service companies that tightly controlling workers' tasks in such a complex environment is necessary to ensure an efficient and error-free service.

However, ODRP drivers operate under significant spatiotemporal pressures and engage in real-time interactions with passengers and dynamic traffic conditions, which means static algorithmic directions often fail to adapt to sudden detours, congestion, or unexpected situations. This creates a unique tension between algorithmic efficiency and the need for human judgment, which is less pronounced in other work environments. Most of our participants had previously worked in the transportation industry or were familiar with the service areas as local residents, enabling them to adapt to challenging tasks and exercise sound judgment in sudden situations. In work environments where many unpredictable external variables occur, flexible work practices that leverage human capabilities can be more effective [8]. As gig workers apply experiential knowledge, learned strategies, and improvisational tactics to their work [37, 49], integrating ODRP drivers' human insight and situational awareness into centralized algorithmic systems could improve work practices, especially in environments with unpredictable external conditions. In our findings, participants believed their abilities could complement algorithmic management and enable more flexible task performance, yet they reported a tension in which their expertise and situational knowledge were frequently overlooked or underutilized in current practices. This tension—between algorithmic mandates prioritizing operational efficiency and human expertise—raises important questions about how algorithmic management can be recalibrated to grant workers autonomy without undermining operational goals. Critically, ODRP drivers' real-time judgment should not be viewed as mere exceptions to algorithmic compliance but as integral contributions that can inform system improvements over time or address unexpected situations. These over-time improvements are especially critical

for employees—such as ODRP drivers—rather than short-term gig workers, given their ongoing engagement and potential for sustained contribution. Moreover, enabling workers to make contributions not only strengthens their sense of autonomy but also serves as a powerful catalyst for intrinsic motivation [7, 33]. Therefore, considering the multitude of unpredictable external variables and the employee status of ODRP workers, it is particularly crucial—more so than in other algorithmically managed work environments—to design algorithmic work directions that collaborate with human capabilities. One promising approach is the development of hybrid interfaces that facilitate collaboration between humans and algorithms rather than a unidirectional flow of instructions.

In addition, evaluating human workers using the same criteria as those used for controlled workplaces poses challenges for workers. When setting evaluation criteria for complex work environments like ODRP, we can consider whether workers have appropriately utilized their unique human capabilities, such as how well they handle unexpected situations or provide satisfactory customer service, rather than relying solely on quantitative criteria like task compliance rates. These non-quantifiable, intangible capabilities can be more effective, particularly in service work that interacts with end users [79], and make service users feel properly cared for and comfortable [13]. By differentiating these evaluations and acknowledging the unique spatiotemporal complexities of ODRP services, we contribute novel insights to broader algorithmic management discourses. Future research is needed to explore appropriate situations for incorporating the capabilities of human workers in ODRP services and how algorithmic systems can direct and oversee workers while respecting their capabilities.

In sum, we highlight the specific challenges of ODRP services, which expose the limitations of purely top-down algorithmic management. We advocate for hybrid, flexible algorithmic systems that treat human autonomy as an asset—one that not only addresses unpredictable environmental factors but also enriches the service experience for both drivers and passengers. This perspective ultimately advances discussions on algorithmic management by demonstrating how worker-centered interventions can serve as catalysts for more adaptive and inclusive algorithmic systems in complex, dynamic service domains.

5.2 Contextualizing Algorithmic Oversight for Adaptive Driver Support

We observed that when participants were unable to follow algorithmic directions due to unavoidable circumstances, the current algorithmic systems transmitted only numerical data to field managers without capturing the situational context. This limitation often resulted in misunderstandings by field managers and diminished trust in algorithmic oversight among participants. In some cases, participants resorted to taking pictures of their surroundings and the system's directions (e.g., GPS screens) to provide evidence for their explanations. However, performing such tasks while driving is cumbersome and dangerous. In this context, the ODRP workplace is characterized by a hybrid system where workers' performance is evaluated both by an algorithmic system and by supervision from a field manager, as seen in traditional workplaces. Unlike gig workers, ODRP drivers, as employees formally employed by the

company, face not only rigid work practices but also heightened concerns about job security. Therefore, in this work environment, the situational context of tasks under algorithmic oversight holds greater significance.

To address this issue, we suggest that ODRP services identify and categorize situations where workers cannot follow algorithmic directions. For instance, it may be possible to design a real-time system to collect field data (e.g., cases where drivers frequently deviate from the assigned routes) from the frontline workplaces. Also, accumulating such data could enable the system to proactively detect problematic situations encountered by workers and offer alternative solutions. For example, the system could adjust task directions that are challenging for workers to follow, as well as the evaluation criteria for such tasks. By incorporating contextual data into the algorithmic systems, ODRP services can better support workers in managing unexpected situations that arise during driving.

Moreover, we found that workers currently have limited channels for voicing their explanations regarding subtle issues that only they experience. To overcome these limitations, we suggest a system that conveys situational data without requiring workers to gather evidence manually. For example, when a road closure forces a driver to deviate from the assigned route, the system could automatically transmit GPS data and dashcam footage to the field manager in real-time. Beyond automated solutions, we can imagine that workers could also provide direct feedback through in-vehicle interfaces explaining why they could not follow specific instructions. Given the unpredictable variables inherent in the ODRP work environment, solely relying on automated data transmission may lead to misunderstandings. Therefore, supplementing automated systems with direct feedback from workers is crucial. For example, considering the workload during driving, workers could submit situational data using simple predefined options (e.g., '*road closure*,' '*traffic congestion*,' '*risk of accidents*'). Such practices, which allow workers to actively participate rather than passively be evaluated, can foster a sense of agency within the algorithmic systems [46]. This sense of agency may be particularly beneficial for employees like ODRP drivers, who must comply with dispatching algorithms, compared to gig workers who retain the option to refuse tasks.

5.3 Disentangling Human Performance from Algorithmic Decisions

We observed that participants were required to follow algorithmic directions while driving; however, most passengers, as the service users, were unaware of this fact. This lack of transparency in the work practices of ODRP services sometimes led to misunderstandings among passengers, causing emotional distress for participants. Unlike traditional workplaces or gig work, where work processes may be relatively easily understood, the ODRP work environment involves real-time routing and dynamics within multiple passengers, intensifying the impact of such misattributions.

To address this issue, we suggest that ODRP services provide passengers with a clear explanation of the roles of algorithmic systems and drivers. For example, during the service onboarding process, the role of the algorithmic systems could be explicitly outlined to enhance passengers' understanding and set their expectations

for how the service operates. A critical implication here is that transparency should not only be informative but also contextually relevant to multi-passenger driving scenarios, clarifying how algorithmic systems determine routing and scheduling, and how passengers' inputs may affect these decisions. Effectively conveying the role and operational principles of the algorithmic system to users of such complex services in a clear and accessible manner can enhance their understanding of the service's operations and foster greater trust [24, 39, 45]. Additionally, in cases where delays occur due to route changes by algorithms, passengers could be informed that the algorithms were responsible for such real-time adjustments. For instance, the service app could proactively notify passengers with a message like, *"The algorithmic system has optimized the routes to ensure that all passengers can reach their destinations more efficiently and quickly."* This approach shifts accountability from drivers to the algorithmic system coordinating ride-pooling through a communication strategy uniquely tailored to the spatiotemporal complexities of the ODRP service. Future research could investigate communication strategies to effectively convey algorithmic decisions and manage passenger expectations in dynamic, multi-passenger ODRP scenarios.

In the case of *Shucle*, passengers could express their satisfaction with the ride experience only through a star rating system, leading to mixed evaluations of both the algorithmic systems and the drivers. We suggest refining the service evaluation system to better differentiate these roles. For example, after using the service, passengers could be asked to rate separately the responsibilities of human workers (e.g., *"Did the driver maintain an appropriate speed?"*) and those of the algorithmic systems (e.g., *"How much was the delay compared to the estimated time?"*). This design not only clarifies the distinct roles of drivers and algorithms for passengers but also serves as a channel for delivering user feedback to service designers. (Many complaints or feedback directed at drivers by passengers could be more appropriately conveyed to those responsible for designing the algorithmic systems.) By segmenting feedback, the system empowers drivers with more explicit accountability boundaries and informs the development of more responsive algorithmic systems. These benefits are especially significant in service domains where tasks are shared between algorithmic systems and human workers, and the communication channels between users (passengers) and service providers are less direct. Building on previous work that addressed the necessity and strategies for explaining the capacity of algorithmic systems to users [30, 71, 72], further research is needed to explore effective ways of communicating the operation of ODRP services, where human workers are required to execute algorithmic directions.

6 LIMITATIONS AND FUTURE WORK

We acknowledge that this study has some limitations. First, our focus on a single ODRP service—*Shucle*—means that worker experiences under algorithmic management may differ in services with other vehicle scales or management strategies. Moreover, as *Shucle* is an ODRP service operated in South Korea—where hierarchical labour practices intersect with a high rate of technology adoption—the stringent surveillance we observed likely arises from a complex interplay of organisational and cultural factors, including

the comparatively nascent service, individual field managers' leadership styles, and the company's unique corporate culture. Because our data derive exclusively from drivers, we are cautious about attributing these factors definitively. Nevertheless, our findings surface fundamental characteristics of algorithmically managed ODRP work environments that, while grounded in *Shucle*'s specific context, may plausibly manifest in other services. Building on our findings as the "ultimate particular" [60], we hope future research will unpack these organisational mechanisms and examine algorithmic management across diverse cultural and corporate settings. Further research on various ODRP services worldwide could offer broader perspectives and a more comprehensive understanding.

Next, we acknowledge a limitation regarding the demographics of our sample. At the time of our study, no drivers younger than 50 years were available, so our findings predominantly capture the experiences of older workers. This age factor could influence the reported worker practices related to the algorithmic systems. However, this focus specifically provides valuable insights into an often underrepresented perspective in algorithmic management research. In spite of it, in future work, examining the experiences of younger ODRP drivers—who may have different levels of digital literacy, familiarity, and trust in algorithmic systems—could enrich our understanding of worker practices in algorithmically managed environments.

Lastly, our findings recognized that workplaces of ODRP services are complicated systems that should consider interactions among workers and surrounding stakeholders. ODRP workers could be directly or indirectly impacted by various stakeholders in their current work practices. Although we engaged with field managers who closely interact with drivers to gain insights into the ODRP work context, their limited number of people—only one or two managers per region (Sejong, Paju)—posed challenges to formally including them in the analysis. Although we focused on investigating workers' experiences to achieve our research goal as a meaningful starting point, further studies involving other stakeholders may unveil multifaceted aspects of work practices in ODRP services. For example, future work should incorporate the passenger perspective by examining how their expectations and feedback influence algorithmic decisions and driver experiences, thereby advancing our understanding of co-produced algorithmic governance. Furthermore, we aim to collaborate with these stakeholders to develop worker-centered designs that can be practically implemented and applied within the ODRP context. While the design implications in this paper have not yet been formally shared with *Shucle* stakeholders, we plan to engage with field managers and developers to gather their feedback and explore opportunities for practical implementation in future work.

7 CONCLUSION

Our work investigated how algorithmic management affects worker practices in ODRP services. Through interviews with 16 drivers working in *Shucle*, an ODRP service in South Korea, we revealed their experiences and perceptions of algorithmic management in their work environment. Although participants acknowledged the general benefits and necessity of algorithmic management, they reported a lack of autonomy in following algorithmic work directions

and expressed frustration with the dynamics among other stakeholders, which were influenced by algorithmic decision-making. Based on the findings, we discussed design implications for supporting algorithmic workplaces in ODRP services. We hope that this study will inspire future researchers to make a meaningful impact on creating a work environment that promotes worker practice at the forefront.

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